

Fig. 5: Actual-size PCB layout of automatic drip irrigation system

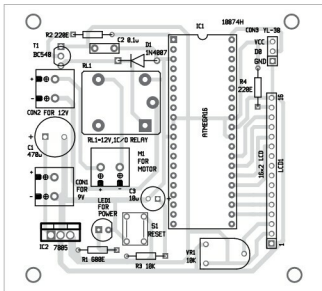


Fig. 6: Components layout for the PCB

it on the PCB. Connect the moisture sensor to the circuit, ensuring that Vcc, GND and D0 pins are properly connected. Note that the moisture sensor includes YL-38 and YL-69 modules. YL-38 is basically a driver and isolator for YL-69 moisture probe (Fig. 8). YL-38 has four pins (Vcc, GND, D0 and A0) on one side, and two pins on the other side, which are used to connect the prongs/probes (YL-69) that are to be inserted into the soil. D0 pin is used for digital output and A0 pin (not used in this project) is used for analogue output. The module also has a potentiometer to calibrate the sensitivity of the sensor.

Connect a 9V battery to the circuit. Regulator IC 7805 provides regulated 5V output to run the circuit. Note that you can also use a 12V DC power supply depending on your motor pump. During testing at EFY Lab, a 12V DC supply was used for the water motor pump.

To test the project, insert the sensor probe into the soil. Switch on the power supply to the circuit. If the soil is dry, the motor will automatically turn on and the LCD will show 'Dry soil, Motor on' message.

You can further extend this

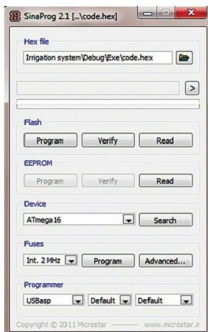


Fig. 7: SinaProg window

project to make a practical drip irrigation system. Typically, a drip irrigation system requires some special



Fig. 8: Prongs with driver

components and control systems. For example, you may need distribution lines (both large- and small-diameter pipes and pipe fittings), electronic or hydraulic control valves, poly fittings and accessories (to make connections), emitters, drippers or spray heads depending on the design and application. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Vasavi Sai VPKD (left) is an editorial member of International Journal of Information Technology. His interests include LED fabrication and designing, embedded systems, home automation, VLSI and FPGA. He has published research papers in international journals. Arup Sarkar (right) is co-founder of briefbay.com website.

